MILITARY SPECIFICATION

MICONAZOLE NITRATE CREAM, TOPICAL

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Miconazole Nitrate Cream, Topical (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Miconazole Nitrate Cream, 2 percent (w/w) within the designated assay limits for the cream.

3.1.1 Shall be suitable for topical use.

3.1.2 pH. The pH of the finished cream shall be not less than 2.50 and not more than 3.50 when determined potentiometrically, undiluted, at 25° C, using the USP method.

3.1.3 Assay. The finished cream shall contain not less than 1.8 percent and not more than 2.2 percent (w/w) of the required amount of miconazole nitrate when determined as specified in 4.8.1.

3.1.4 Identity. The infrared absorption spectrum of a potassium bromide dispersion of the extracted Miconazole Nitrate shall exhibit maxima only at the same wavelengths as that of a similar dispersion of Miconazole Nitrate Reference Standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

3.2 Active ingredients.

3.2.1 Miconazole Nitrate.

3.2.1.1 Loss on drying. Shall not exceed 0.5 percent when dried at 105° C for two hours.

3.2.1.2 Ultraviolet absorption. The ultraviolet absorption spectrum of a solution containing approximately 40 mg in 100 ml of acidic methanol (qs 100 ml N/10 HCl to 1 liter with methanol), accurately weighed, shall exhibit maxima at 265, 272, and 280 nm ± 2 nm, and the respective absorptivities shall not differ by more than 3.0 percent compared to Miconazole Nitrate Reference Standard concomitantly determined.

3.2.1.3 Infrared spectrum. The infrared absorption spectrum of a potassium bromide dispersion of it previously dried at 105° C exhibits maxima only at the same wavelengths as that of a similar preparation of Miconazole Nitrate Reference Standard. Transmittance of the strongest band shall be between 5 - 15 percent.

3.2.1.4 Residue on ignition. Shall not exceed 0.2 percent when determined by the USP method.

3.2.1.5 Foreign related substances. The thin layer chromatograph shall show only one spot, at the approximate Rf value as that of The Reference Standard, when determined as specified in 4.9.1.

3.2.1.6 Assay. Shall contain not less than 98.0 percent and not more than 102.0 percent of Miconazole Nitrate when determined as specified in 4.9.2.

3.2.1.7 Heavy metals. Shall contain no more than 20 ppm when determined by the USP Method II.

3.3 Other ingredients. All other ingredients entering into the manufacture of the cream shall be of USP or NF quality, or, if not included in either of these compendia they shall be suitable for use in the cream.

3.4 Workmanship. The material and its containers shall be free from any defects which detract from their appearance or impair their serviceability.

3.5 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.6 Delivery. Not more than 4 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:10Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform and homogeneous quantity of Miconazole Nitrate cream produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105, with an inspection level of S-2 and an acceptable quality level (AQL) of 1.0 percent defective for major defects.

4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.6 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

TABLE I. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
pH	3.1.2	S-2	5 } Composite
Assay	3.1.3	S-2	5 } Composite
Identity	3.1.4	S-2	5 } Composite
Ingredient: Miconazole Nitrate	3.2.1	2/	

1/ Where possible the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.7 Examination. The finished cream shall be examined to determine compliance with all requirements of this specification . Nonconformance will be permitted to the extent indicated in 4.3, 4.4, 4.5, and 4.6.

4.7.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE II. Classification of defects.

Categories	Defects 1/
Major	
101	Cream not free from foreign matter.
102	Cream not uniform in color.
103	Cream not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

4.8 Tests of finished cream.

4.8.1 Assay.

Reagents

Ethyl Orange Solution - Dissolve 100 mg of ethyl orange in sufficient purified water to make 100 ml.

Solvent Mixture - Mix 250 ml of 2-propanol, 245 ml chloroform, and 5 ml glacial acetic acid, all reagent grade.

Standard solution - Accurately weigh approximately 75 mg of miconazole nitrate standard into a 250 ml volumetric flask. Dissolve in and dilute to volume with solvent mixture.

Procedure - Accurately weigh approximately 1.5 Grams of sample into a 100 ml volumetric flask. Dissolve in and dilute to volume with solvent mixture.

Pipet 10.0 ml of 0.1N Hydrochloric acid, 25.0 ml of chloroform and 1.0 ml of ethyl orange solution in that order into each of 6 cylindrical separatory funnels (60 ml capacity).

Pipet 2.0 ml aliquots of sample solution into one pair of separatory funnels, 2.0 ml aliquots of standard solution into another pair, and 2.0 ml aliquots of solvent mixture into the third pair.

Mechanically shake the separatory funnels for 1/2 hour, permit the layers to separate, and draw off the lower chloroform layer into separate 50 ml volumetric flasks.

Pipet 15.0 ml of chloroform into each of the separatory funnels and shake for 15 minutes. Draw off the lower chloroform layers and add to the respective volumetric flasks. Dilute to volume with chloroform.

Determine the absorbances of the solutions in 1 cm cells, in a suitable spectrophotometer at 430 nm, using chloroform as the blank.

Calculation:

$$\frac{(Au - B) \times \text{mg standard} \times 100}{(As - B) \times 250 \times \text{mg sample}} \times 100 = \text{Percent Miconazole Nitrate (W/W)}$$

Where: Au = absorbance sample
As = absorbance standard
B = absorbance reagent blank solution

4.9 Miconazole Nitrate powder.

4.9.1 Foreign related substances. Prepare a glass silica gel plate by dipping into a solution of 5 grams of NaOH in 250 ml of methanol, and dry.

Source: https://assist.dla.mil -- Downloaded: 202610-21T11:10Z
Check the source to verify that this is the currentversion before use.

Procedure. Dissolve 30 mg in a mixture of 2 ml of chloroform and one ml methanol. Spot 10 ul of this solution on the plate prepared as above. Spot the same amount of miconazole nitrate referenced standard similarly prepared, on the plate adjacent to the sample. Develop the plate using a mixture of 6 volumes of hexane, 3 volumes of chloroform, and one volume of methanol. After drying the plate, place in a chamber containing iodine vapors to visualize the spots.

As an alternate approximately 65 mg of miconazole nitrate sample, accurately weighed, shall be dissolved in, and diluted to 25.0 ml in a solvent mixture consisting of chloroform:methanol in a 2:1 ratio, respectively. Spot 100 ul of the solution and the same amount of miconazole reference standard solution, similarly prepared, onto the alkaline silica gel plate and place into the developing tank containing the mixture of solvents described in the preceeding paragraph. After development, dry, and spray with a solution of 1 percent iodine in chloroform.

4.3.2 Assay. Dissolve approximately 200 mg of dried miconazole nitrate in 50 ml of glacial acetic acid. Add 8 drops of alpha-naphtholbenzein indicator solution (consisting of 80 mg dissolved in 100 ml of glacial acetic acid), and titrate with 0.02N standardized volumetric solution perchloric acid. Perform a blank determination and make any necessary correction. Each ml of 0.02N perchloric acid is equivalent to 9.583 mg of $C_{18}H_{14}Cl_4N_2O.HNO_3$.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One package containing one tube constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available tube considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Six unit packages shall be packaged in an intermediate package as specified in 5.2.12.1 of PPP-C-186.

Source: https://assist.dla.mil -- Downloaded: 202610-21T11:10Z
Check the source to verify that this is the currentversion before use.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twenty-four intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.3.5.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II Shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-01-017-3095	MICONAZOLE NITRATE CREAM, TOPICAL, 2%, 1 oz (28.35 Gram), 12s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1779

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-M-37636 (DSA-DM) MICONAZOLE NITRATE CREAM, TOPICAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

☆ GPO 1974—603-680/6126 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:10Z
Check the source to verify that this is the current version before use.